CERTIFYING THE JUDGE: FALSIFIABLE PROPERTIES FOR LLM-BASED EVALUATION OF FORMAL CODE

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ABSTRACT

LLM judges are increasingly used to evaluate AI-generated code and formal specifications, especially where passing tests or type-checking cannot show that an artifact captures its intended meaning. Practitioners typically establish a judge's reliability through agreement with human ratings and must repeat that check for every new judge, prompt, model, and domain. We ask whether one human panel can instead be amortized through falsifiable behavioral properties that any new judge is tested on automatically. We propose a property-certification protocol for LLM judges of reference-based Lean 4 specification quality: how completely and correctly a candidate specification captures a gold reference. Each judge is tested for identity, monotonicity under injected bugs, monotonicity under removed specifications, and repeat/prompt stability. A variance-aware trust index, $\mathrm{TI}_{\mathrm{var}}$, summarizes bug monotonicity and score stability with fixed equal weights. Across three judge families with eight configurations each, including weaker GPT-4o-mini judges, a lexical grader, an edit-distance scorer, and a random control, $\mathrm{TI}_{\mathrm{var}}$ is positively associated with judge-level human agreement on a 72-pair development panel. For Codex judges, the Spearman correlation is 0.833 on averaged labels and 0.905 and 0.738 on the two rating passes. Specification monotonicity is the most informative single property, and the index flags a lexical judge whose human agreement alone looks acceptable. Applied to VERIBENCH, an LLM judge ranks agent-generation methods in the expected order without new labels. Judge validation thus moves from a per-judge annotation cost toward a one-time human panel plus automated, falsifiable checks that any new judge must pass.

1 INTRODUCTION

Agentic AI systems increasingly generate both code and the checks meant to establish trust in it, such as unit tests (Haque et al., 2026) and formal specifications (Yang et al., 2024; Bursuc et al., 2025). Deciding whether such a check is itself correct is the test-oracle problem (Barr et al., 2015), now posed for checks that an AI wrote. When the same system writes the code and its checks, the checks can be gamed: coding agents modify or delete failing tests instead of fixing the code (Baker et al., 2025; Zhong et al., 2025). Running a check cannot settle whether it is right, because a passing test shows only that the code and the test agree. Benchmarks of AI-written tests accordingly grade them against a human-written reference: a generated test counts as reproducing an issue only if it fails before and passes after the developers' fix (Mündler et al., 2024). Writing a formal specification is the formal analogue of writing the tests that SWE-bench uses to grade a patch (Jimenez et al., 2024), with a stronger guarantee. A property-based test checks a property on sampled inputs (Claessen & Hughes, 2000), whereas a theorem asserts it for every input.

In Lean 4, these artifacts are code-adjacent formal objects: runnable programs paired with unit tests, formal specifications (theorem statements about the program), and, sometimes, proofs of those theorems. In classical autoformalization, the target theorem is supplied, so evaluation can often reduce to proof success. In agentic verification, the system must decide *what should be true* before it can prove anything. The Lean kernel can check a proof, but it cannot tell us whether the proposed theorem is complete, semantically aligned with the intended program, or merely a weak statement that happens to typecheck (Lahiri, 2024; Yang et al., 2024). No kernel check or test run supplies this judgment of intent, so it falls to a human oracle (Barr et al., 2015): autoformalization benchmarks rely on expert review of statement correctness (Azerbayev et al., 2023). As agents produce more candidate

artifacts than humans can inspect, specification-quality evaluation becomes a scalable-oversight problem (Bowman et al., 2022; Zhou et al., 2025). LLM judges already stand in for the human oracle at scale. Open theorem-prover pipelines use them to filter over a million autoformalized Lean statements (Ying et al., 2024; Lin et al., 2025; Wang et al., 2025), and they are increasingly used to evaluate AI-generated code (Zhuo, 2024; Tong & Zhang, 2024; He et al., 2025). Yet the reliability of these judges on formal statements has received little study (Chen et al., 2026b). This paper asks when such a judge of AI-written formal specifications can be trusted.

The standard reliability check is per-judge human validation: collect labels, measure agreement between each judge and human ratings, and keep the judge if the correlation is high enough (Liu et al., 2023; Zheng et al., 2023). The check is necessary because judge–human agreement varies substantially across judge models, datasets, and evaluated properties (Bavaresco et al., 2024). Its cost, however, grows with every new judge, prompt, model update, and domain, and each one asks for another round of expert annotation.

We instead collect human labels once and test each new judge automatically against falsifiable properties whose correct direction is known by construction. A trustworthy judge should give the maximum score to an artifact compared with itself, and lower scores as semantic bugs accumulate or as theorem statements and unit tests are removed. Its scores should also stay stable across repeated calls and semantics-preserving rewrites. The one-time human panel measures how strongly these properties track human agreement; afterwards, a new judge configuration is screened by running the property suite, with no new labels.

On a 72-pair development panel with eight judge configurations, our variance-aware trust index $\mathrm{TI}_{\mathrm{var}}$ is associated with judge-level human agreement (Spearman 0.833 on averaged labels for Codex judges), and the association is positive for Codex, Claude Opus, and Qwen3-Coder judges alike. Human agreement alone can miss a bad judge: a text-similarity grader agrees with humans about as well as properly prompted judges, yet $\mathrm{TI}_{\mathrm{var}}$ ranks it below all of them. These are development-panel associations; a locked held-out test with independent raters remains future work.

Our **contributions** are:

1. **An amortized validation protocol:** a one-time human panel plus four falsifiable properties lets any new (model, prompt, repeats, aggregation) judge configuration be screened without new labels (Section 4).
2. **Development-panel evidence:** across three judge families, $\mathrm{TI}_{\mathrm{var}}$ is associated with judge-level human agreement, and specification monotonicity is the most informative single property (Section 5.3).
3. **Downstream use:** an LLM judge ranks VERIBENCH specification-generation agents in the expected order, as a downstream sanity check (Section 5.4).

2 RELATED WORK

LLM-as-a-judge. LLM judges are validated mainly by agreement with people: G-EVAL reports Spearman correlation with human scores (Liu et al., 2023), MT-Bench reports agreement with human preferences (Zheng et al., 2023), and Judge-Bench tests whether LLMs replicate human annotations across 20 NLP datasets (Bavaresco et al., 2024). These audits find that agreement varies by task and setup, and that judges show position and verbosity biases and sensitivity to prompt phrasing (Bavaresco et al., 2024; Schroeder & Wood-Doughty, 2024). Multi-agent debate (Chan et al., 2024), rubric-driven prompting (Liu et al., 2023), and internal-consistency measures over repeated scorings (Schroeder & Wood-Doughty, 2024) improve or quantify reliability, and simple comparative prompting can grade more consistently than long reasoning chains (Chen et al., 2026a). We keep human agreement as the validation target but measure it once, then screen later judges with falsifiable properties.

Judges for code. For code, pass@k is unavailable when tests are missing and misleading when they are weak; ICE-SCORE and CODEJUDGE instead prompt LLMs to judge functional correctness (Zhuo, 2024; Tong & Zhang, 2024). Our properties resemble metamorphic testing: each perturbation has a known effect on the correct score, so a judge's response to it can be checked without new labels.

Judging formal statements. Autoformalization work usually supplies the target theorem and scores proof success, whereas agentic verification systems must also propose the specification (Miranda et al., 2026; Bursuc et al., 2025). Checking that a proposed statement is faithful increasingly relies on models: FORMALALIGN scores the alignment between informal and formal statements (Lu et al., 2024), and open theorem-prover pipelines filter autoformalized statements with LLM judges (Lin et al., 2025; Wang et al., 2025). ReForm validates such judges once, against 859 expert-annotated items, and finds that even experts make semantic errors in up to 38.5% of cases (Chen et al., 2026b). Debate can help weaker judges supervise stronger models (Kenton et al., 2024); our setting instead has gold references, which the properties exploit to test new judges without new labels.

3 PROBLEM SETUP: JUDGING LEAN FORMAL-SPECIFICATION QUALITY

Task. Each benchmark item contains a natural-language problem statement, a gold Lean 4 artifact g, and a candidate artifact x produced by an AI agent. We call the informal requirement conveyed by the problem statement the *intent*. An artifact is a single Lean file containing an implementation, a *formal specification* of it, and possibly unit tests and proofs. Following common formal-methods usage, the formal specification is the set of theorem statements about the implementation, together with the predicates they use. Typical examples are pre/postconditions, algebraic properties, $\mathrm{Pre} \to \mathrm{Post}$ correctness, and equivalence to a reference implementation. Unit tests are not part of it, and proofs only certify that its theorems hold; Appendix F discusses the competing view that a unit test is a ground theorem. Hereafter, "specification" means formal specification, whether an agent, a human, or a script wrote it. Our goal is to assign a score $J(x, g) \in [0, 1]$ of *reference-based specification quality*: how completely and correctly the candidate captures the intended formal specification represented by the gold artifact.

Why this is nontrivial. Type-checking does not determine whether x formalizes the same intended property as g: a candidate may compile while being under-specified, over-constrained, vacuously true, or semantically shifted. In VERIBENCH, this often appears as a coverage gap, where a candidate captures only part of the behavior the gold reference specifies (Miranda et al., 2026).

Why a gold reference. Without ground truth, direct LLM ratings are vulnerable to strategic manipulation, which motivates information-theoretic alternatives to direct rating queries (Robertson & Koyejo, 2025). Our setting has ground truth in the form of an audited gold reference. Scoring against it ties each rating to a question with a correct answer in principle: which of the reference's guarantees the candidate states. The score therefore targets coverage and correctness relative to g, not an unanchored preference.

Judge model. We use an LLM judge parameterized by

$$\theta = (\text{model}, \text{prompt}, n_{\text{repeats}}, \text{aggregation}),$$

which maps a pair (x, g) to a score $J_\theta(x, g) \in [0, 1]$. Section 4 asks when J_θ is trustworthy enough to rank systems without new human labels for every θ.

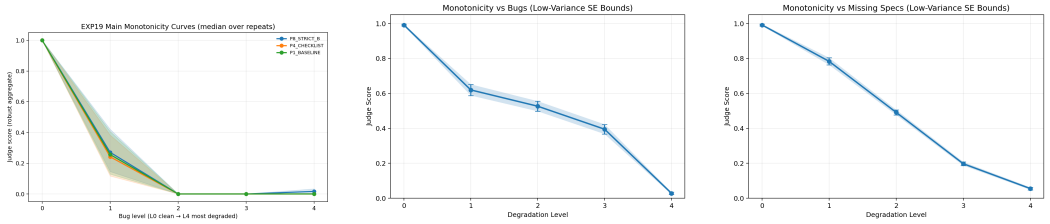
4 PROPERTY CERTIFICATION PROTOCOL

One-time human panel. Humans label a fixed development panel once; in our study, one annotator rated candidate–gold pairs under two rubric strictness levels (Section 5.1). For each judge configuration, we then compute property scores on scripted perturbations of Lean artifacts, which requires judge calls but no human labels. The panel is used only to measure how strongly property scores track human agreement (Table 1).

Properties. A trustworthy judge should satisfy four falsifiable behavioral targets. **P1 (identity):** an artifact compared with itself receives the maximum score, $J(x, x) = 1$. **P2 (bug monotonicity):** scores decrease as semantic bugs are introduced cumulatively. **P3 (specification monotonicity):** scores decrease as theorem statements and unit tests are removed; we keep this historical name although, by the definition in Section 3, removed tests are not specification. **P4 (stability):** scores stay stable across repeated calls and under semantics-preserving rewrites such as ALPHA-renaming,

Step	Action
1	Collect human ratings of candidate-vs.-gold Lean artifacts once (the development panel).
2	For each judge configuration, score scripted perturbation ladders and compute identity, bug monotonicity, specification monotonicity, and stability; this needs judge calls, not human labels.
3	Aggregate the property scores into a trust index such as $\mathrm{TI}_{\mathrm{var}}$.
4	On the panel, measure how well the index tracks human agreement; afterwards, compare new configurations by the index alone.

Table 1: **Property-certification workflow.** Human labels enter once (steps 1 and 4); later judge configurations need only steps 2 and 3.



(a) Low-variance monotonicity (main view)

(b) P2: Monotonicity vs. Bugs (Mean ± SE)

(c) P3: Monotonicity vs. Missing Specs (Mean ± SE)

Figure 1: **Property sanity checks for a GPT-5.3-Codex judge** on VERIBENCH ($n_{\text{files}} = 184$, $n_{\text{repeats}} = 3$). **(a)** Low-variance view of scores under controlled degradations. **(b)** P2: scores fall as bugs accumulate, with local reversals (Spearman $\rho = -0.808$; weak monotonicity 52.2%). **(c)** P3: scores fall more consistently as theorem statements and unit tests are removed ($\rho = -0.899$; weak monotonicity 89.1%). Table 2 reports P4 rewrite invariance.

which renames bound variables and binders without changing what the artifact means. We measure monotonicity with Kendall's τ, Spearman's ρ, and pairwise ordering accuracy, following prior judge-evaluation work (Tong & Zhang, 2024; Zhuo, 2024).

Trust indices. The property scores ident (P1), mono (P2), weak (P3), and stab (P4) lie in $[0, 1]$ and are graded targets, not binary gates. As baselines, the arithmetic indices $v1$ and $v2$ take fixed weighted averages of them, and GM aggregates monotonicity and stability geometrically (Appendix I.5). Our primary index, $\mathrm{TI}_{\mathrm{var}}$, is the geometric mean of a bug-monotonicity term and two stability terms, with fixed equal weights:

$$\mathrm{TI}_{\mathrm{var}} = \exp\Big(\tfrac{1}{3}\log C_f + \tfrac{1}{3}\log(1 - \widetilde{\mathrm{VarCorr}}) + \tfrac{1}{3}\log(1 - \widetilde{\mathrm{VarScore}})\Big),$$

where C_f is the mean per-file bug-ladder monotonicity term and $\widetilde{\mathrm{VarCorr}}$, $\widetilde{\mathrm{VarScore}}$ are normalized variance penalties (Appendix I.5). Because the index is multiplicative, one weak term pulls it down, and the variance penalties favor judges that behave consistently across repeats and prompts. Our hypothesis is that judges with higher $\mathrm{TI}_{\mathrm{var}}$ agree more with humans on the development panel.

5 EXPERIMENTS

5.1 DATA AND HUMAN PANEL

All experiments use VERIBENCH (Miranda et al., 2026), whose 857 theorems span five subsets: introductory logic and programming tasks (Easy), classical data structures and algorithms (CS), HumanEval puzzles (Chen et al., 2021), security labs on real code (Security), and Python standard-library programs (RealCode). The development panel contains 75 candidate–gold pairs (25 problems, three agent generations each), each rated on a 0–5 rubric by one annotator under two rubric strictness levels, pass1 and pass2. Because one problem's gold file is missing, 72 pairs from 24 problems enter the analyses; we report pass1, pass2, and their average. The two passes agree only moderately

Rewrite	Mean Score	Mean Within-Item Std
RENAME	0.936	0.005
REORDER	0.849	0.029
STYLE	0.982	0.001
ALPHA	0.399	0.096

Table 2: **P4 semantic invariance for the Codex judge.** Scores under semantics-preserving rewrites; ALPHA-renaming is the weakest mode.

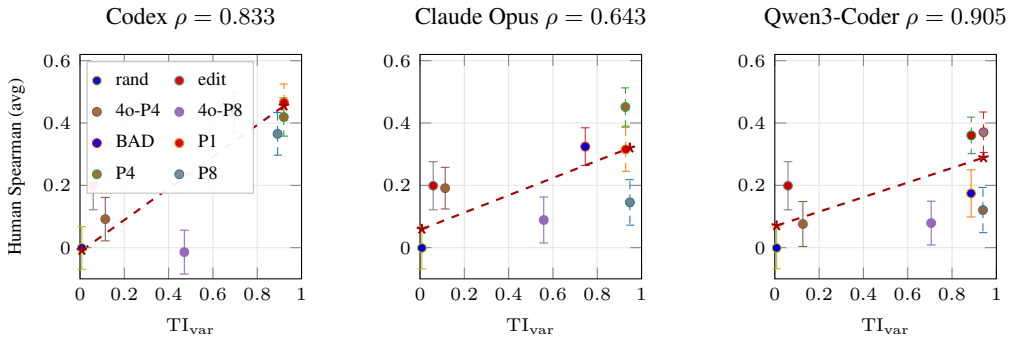


Figure 2: $\mathrm{TI}_{\mathrm{var}}$ **vs. judge-level human agreement (averaged labels).** One panel per judge family and one point per configuration; the y-axis is Spearman agreement with averaged human labels on the development panel, with approximate standard errors. Panel titles give the judge-level Spearman ρ reported in Table 3; dashed lines are OLS guides only. An expanded 13-judge panel is shown in Appendix Figure 7.

(Spearman 0.249, quadratic-weighted $\kappa = 0.372$, MAE 0.778; Table 10), which bounds any judge–human correlation well below 1. The same panel is reused for every judge, so differences between judges reflect judge behavior, not row selection.

5.2 PROPERTY SANITY CHECKS ON GPT-5.3-CODEX

Figure 1 shows the intended qualitative behavior: scores fall as bugs accumulate and as theorem statements and unit tests are removed. Specification monotonicity is the cleaner signal ($\rho = -0.899$); bug monotonicity is directionally correct but noisier ($\rho = -0.808$), with local reversals. Claude Opus judges show the same pattern (Appendix Figures 5 and 6). Table 2 exposes the main failure mode: scores barely move under renaming, reordering, and style edits, but drop to 0.399 under ALPHA-renaming, which leaves Lean semantics unchanged (see also Appendix Figure 12).

Why ALPHA-renaming is hard. ALPHA-renaming is the hardest rewrite for our judges even though it leaves Lean semantics unchanged. Renamed binders move an artifact away from both the gold reference and typical training text, so lexical cues and naming conventions can shift scores even when equivalence is obvious to an expert. The perturbation supplies no certificate of α-equivalence, so the judge must infer invariance from surface patterns, which current models do only partly. Canonicalizing binders or giving the judge kernel-derived structure are natural fixes.

5.3 $\mathrm{TI}_{\mathrm{var}}$ TRACKS JUDGE-LEVEL HUMAN AGREEMENT ON THE DEVELOPMENT PANEL

Each judge family (Codex, Claude Opus, Qwen3-Coder) has eight configurations: three prompts (P1, P4, P8), two GPT-4o-mini configurations, a text-similarity grader (BAD_STRINGY), an edit-distance scorer, and a random control. Figure 2 plots each configuration's $\mathrm{TI}_{\mathrm{var}}$ against its Spearman agreement with averaged human labels. The judge-level association between $\mathrm{TI}_{\mathrm{var}}$ and human agreement is positive in every family (Table 3). Six of the nine family–pass correlations exceed 0.7, and Claude Opus, the weakest family, stays positive on every pass. For Codex, Pearson correlation shows the same pattern (avg $r = 0.774$), while Kendall's τ is noisier at $n = 8$ (avg $\tau = 0.571$).

	Averaged labels				$\mathrm{TI}_{\mathrm{var}}$ by pass	
Judge family	$v1$	$v2$	GM	$\mathrm{TI}_{\mathrm{var}}$	pass1	pass2
Codex	0.762	0.476	0.857	0.833	0.905	0.738
Claude Opus	0.429	0.143	0.714	0.643	0.810	0.595
Qwen3-Coder	0.524	0.500	0.786	0.905	0.476	0.905

Table 3: **Judge-level association between trust indices and human agreement** (development panel, eight configurations per family). Each entry is the Spearman correlation between an index and the configurations' Spearman agreement with human labels. $v1$, $v2$: arithmetic indices; GM: geometric index; $\mathrm{TI}_{\mathrm{var}}$: variance-aware index.

Codex-family configuration	**Human ρ (avg)**	$\mathrm{TI}_{\mathrm{var}}$	**Bug-ladder reversal rate**
P1 prompt	0.47	0.92	0.00
P4 prompt	0.42	0.92	0.00
P8 prompt	0.37	0.89	0.17
BAD_STRINGY (lexical)	0.43	0.70	0.83
GPT-4o-mini, P8 prompt	−0.01	0.47	0.75
GPT-4o-mini, P4 prompt	0.09	0.11	0.67
Edit distance	0.20	0.06	1.00
Random	0.00	0.01	1.00

Table 4: **Codex-family configurations.** Human agreement and $\mathrm{TI}_{\mathrm{var}}$ are the values plotted in Figure 2; the reversal rate is the fraction of 24 bug ladders with at least one local score increase (Appendix I). The lexical grader matches the prompted judges on human agreement but not on the properties.

Human agreement alone can miss a bad judge. BAD_STRINGY is a deliberately lexical grader. On the Codex panel it agrees with humans about as well as the three prompted judges (Table 4), consistent with the panel labels partly tracking surface similarity to the gold reference: edit distance alone reaches 0.20. Yet its scores reverse on 20 of 24 bug ladders, and $\mathrm{TI}_{\mathrm{var}}$ ranks it below all three prompted judges. Screening on human agreement alone would have kept this judge; the property suite exposes it.

Choice of index. $\mathrm{TI}_{\mathrm{var}}$ improves over the arithmetic $v2$ in every family (Codex $0.476 \to 0.833$, Claude Opus $0.143 \to 0.643$, Qwen3-Coder $0.500 \to 0.905$; Table 3). The plain geometric index is competitive for Codex and Claude Opus (0.857 and 0.714); we adopt $\mathrm{TI}_{\mathrm{var}}$ because its variance penalties target instability directly and it is positive for every family and pass. A separate weight-tuned geometric variant, fit on one label pass, keeps Spearman 0.786–0.810 on the other pass over the same pairs (Appendix Table 7). On an expanded 13-configuration panel, its pass1-fit form (mono$^{0.4}$ stab$^{0.6}$) reaches $\rho = 0.769$ with a bootstrap 95% interval of $[0.34, 0.95]$ (Appendix Figure 7). Intervals this wide mean that, with eight to thirteen configurations, these correlations establish direction and rough strength rather than precise effect sizes. Because the index form was chosen with these labels in view, all of these numbers are development-panel associations, not held-out estimates.

Which properties matter. Table 5 correlates each property with human agreement across the eight Codex-family configurations. Specification monotonicity (P3) has the strongest association on every label view (Spearman 0.63–0.78), followed by bug monotonicity (P2; 0.60–0.71). Identity (P1) and stability (P4) are weak or inconsistent standalone signals. With eight configurations, only P3 on pass2 and on averaged labels reaches permutation $p < 0.05$.

Decision threshold. To turn scores into accept/reject decisions, we sweep thresholds $t \in \{0.60, 0.65, \ldots, 0.85\}$. Bug-ladder ranking is unchanged ($\rho_{\text{bug}} = -0.790$ at every t), while stricter thresholds slightly lower the rate at which repeated calls flip a decision (0.058 at $t = 0.60$ to 0.046 at $t = 0.85$) and lower the pass rate (0.583 to 0.536; Appendix Table 8). We select $t = 0.85$, which has the lowest observed flip rate.

Property	pass1	pass2	avg
P1 identity	−0.21 (0.62)	0.02 (0.98)	−0.02 (0.98)
P2 bug monotonicity	0.60 (0.13)	0.67 (0.08)	0.71 (0.06)
P3 specification monotonicity	**0.63** (0.11)	**0.78** (0.03)	**0.75** (0.04)
P4 stability	0.47 (0.24)	−0.23 (0.57)	0.14 (0.76)

Table 5: **Per-property association with human agreement** (eight Codex-family configurations). Spearman correlation between each property score and the configurations' human agreement; permutation p-values in parentheses.

Agent	Easy	CS	Real	HE	Sec	Avg
Baseline	0.580	0.690	0.472	0.410	0.347	**0.470**
DSPy ReAct	0.659	0.754	0.580	0.650	0.454	**0.615**
Trace+	0.342	0.680	0.592	0.573	0.411	**0.477**
Trace++	0.620	0.756	0.504	0.645	0.432	**0.588**

Table 6: **Specification quality of four agents on VeriBench**, by split (HE: HumanEval; Sec: Security). Claude 3.7 is both the agent model and the judge; the judge returns integer scores in $\{0, \ldots, 10\}$, which we divide by 10 and average over five calls and all items. DSPy ReAct and Trace+ self-debug; Trace++ also self-improves.

5.4 Downstream use: ranking VeriBench agents

Table 6 uses a Claude 3.7 judge to score specifications generated by four agents. Plain prompting scores lowest (0.470), and feedback-driven agents score higher, led by DSPy ReAct (0.615) and Trace++ (0.588) (Khattab et al., 2023; Cheng et al., 2024). This is a sanity check, not independent evidence of judge correctness, and the Claude 3.7 judge is not among the configurations analyzed in Section 5.3.

6 Limitations and Held-Out Test Protocol

Development panel, not a held-out test. The same human labels informed the index form, repeat aggregation, threshold, and prompt panel, so Section 5.3 reports development-panel associations, not out-of-sample estimates. The panel is small (72 analyzed pairs, one annotator, eight configurations per family), and its two rating passes agree only moderately. We also collected 343 labels from five raters over 78 candidate–gold pairs, which the development-panel results do not use. That pool was rated under a different rubric (statement coverage with a spam deduction) and correlates only weakly with the panel labels (Spearman ≈ 0.13 on 63 shared pairs), so it measures a different criterion rather than offering a held-out test of the same one. A locked held-out test therefore needs independent raters under one frozen rubric, with the properties and $\mathrm{TI}_{\mathrm{var}}$ fixed in advance.

What certification does not cover. No finite perturbation suite can rule out every judge failure, so $\mathrm{TI}_{\mathrm{var}}$ is a graded trust signal, not a pass/fail certificate. Scores are specification-relative: they measure agreement with an audited but necessarily incomplete gold reference, since no algorithm decides specification equivalence in general (Rice, 1953). Scores are anchored to a gold reference, but we do not evaluate candidates optimized against the judge, so resistance to gaming remains untested. Neither the judge prompts nor the development-panel rating protocol explicitly addresses presentation aspects of specification quality, such as redundant theorem statements, oversized implementations, or needless abstraction; only the held-out reviewer pool's rubric deducts for superfluous theorem statements. Holistic panel ratings may reflect these aspects implicitly, but we did not measure this; future rating protocols should make them explicit.

7 Conclusion

We proposed an amortized protocol for validating LLM judges of formal specifications: collect human labels once, then screen each new judge with falsifiable properties whose correct direction is

known by construction. On a 72-pair development panel, a variance-aware trust index built from bug monotonicity and score stability is positively associated with judge-level human agreement in all three judge families, and it flags a lexical judge that human agreement alone would keep. Specification monotonicity carries the strongest signal. The key next step is a locked held-out test with independent annotators under one frozen rubric.

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A Additional Development-Panel Results

B Related Work (Cont.)

Autoformalization. Complementing this line of work, Wang et al. (2024a) present TheoremLlama, a framework aimed at enhancing LLM translation into Lean 4. Drawing on over 100K proof examples from the Mathlib4 library, TheoremLlama employs a novel natural-language-to-formal-language (NL-FL) bootstrapping strategy and iterative proof synthesis. This enables the reuse of verified examples as templates for future translations. The framework achieves 36.48% and 33.61% accuracy on the MiniF2F-Valid and MiniF2F-Test benchmarks, respectively—surpassing GPT-4 by more than ten percentage points on both.

Protocol	Fit Spearman	Test Spearman
pass1 → pass2	0.881	0.810
pass2 → pass1	0.857	0.786

Table 7: **Cross-pass robustness of a weight-tuned geometric variant** (not $\mathrm{TI}_{\mathrm{var}}$). Weights over six property scores are tuned on one pass by random search; *Fit Spearman* is the in-sample judge-level Spearman between the tuned index and human agreement on that pass, and *Test Spearman* is the same statistic on the other pass. Both passes rate the same n=75 pairs, so the test pass checks robustness to a second label pass, not generalization to new items or raters.

Threshold	Flip Rate (95% CI)	Pass Rate (95% CI)
0.60	0.058 [0.048, 0.069]	0.583 [0.559, 0.608]
0.70	0.049 [0.040, 0.059]	0.559 [0.537, 0.581]
0.80	0.051 [0.042, 0.061]	0.544 [0.523, 0.566]
0.85	**0.046** [**0.037**, **0.057**]	0.536 [0.515, 0.557]

Table 8: **Threshold selection.** Stricter thresholds slightly reduce decision flips across repeated calls at a moderate cost in pass rate; bug-ladder ranking is unchanged ($\rho_{\mathrm{bug}} = -0.790$ at every threshold).

Techniques for Code Verification. IMPROVER (Ahuja et al., 2024) introduces a Lean-aware Chain-of-States prompting loop that integrates retrieval, best-of-n sampling, and iterative correction to rewrite formal proofs with improved properties. By optimizing for metrics such as brevity and readability, ImProver reduces the number of tactics by half, doubles proof readability, and boosts theorem prover acceptance rates by over 80%. In addition, CLOVER (Sun et al., 2024) implements a closed-loop pipeline in which an LLM first generates code, docstrings, and formal annotations, then uses reconstruction-based prompting to enforce consistency across these outputs, and finally applies SMT-based verification to validate correctness. Evaluated on the CloverBench suite of Dafny programs, Clover accepts 87% of correct solutions, rejects 100% of flawed ones, and even uncovers bugs in human-written code—demonstrating the power of hybrid generation-verification pipelines. In a complementary direction, Zhou et al. (2025) improve general-purpose LLM-based graders by augmenting them with "privileged" information such as gold-standard solutions, grading rubrics, and detailed annotations. When necessary, the system provides targeted hints back to candidate models. This approach achieves grading performance on par with or exceeding that of specialized systems—and even expert humans—on difficult programming benchmarks.

Agentic Frameworks and Tools for Code Verification. TRACE (Cheng et al., 2024) proposes *generative optimization*, tuning entire computational workflows—including code, prompts, tool calls, and error signals—by treating execution traces as gradients in the OPTO framework. With a PyTorch-like API and the LLM-based optimizer OptoPrime, it supports diverse tasks such as prompt tuning, debugging, and robot control, rivaling specialized optimizers. Building on modular composition, DSPY (Khattab et al., 2023) treats LLM calls as declarative modules in a computational graph. Users define concise input–output signatures, and DSPy's compiler automatically bootstraps or fine-tunes pipelines using built-in "teleprompters." This enables a few-line programs to outperform expert-crafted prompts in math, QA, and agent workflows. Extending agentic capabilities to formal reasoning, PANTOGRAPH (Aniva et al., 2024) offers a programmatic interface to Lean 4 with support for advanced proof search. It exposes internal proof states and tactics for integration with learning agents, replacing human-facing interfaces with API-level control.

B.1 RELATED WORK: SCALABLE OVERSIGHT

Scalable Oversight with LLM Judges Researchers have proposed multi-agent oversight frameworks such as debate and consultancy, where weaker LLMs act as judges supervising stronger agents (Kenton et al., 2024; Chan et al., 2024). Debate, in which two models argue opposing sides before a judge, consistently improves truth-tracking compared to single-pass QA. Kenton et al. (2024) demonstrate that debate reduces the likelihood of judges being misled in math and code tasks, highlighting it as a scalable oversight mechanism.

Self-Improving Critics. In the absence of stronger supervisors, models can bootstrap judgment skill through self-critique. The Self-Evolving Critic (SCRIT) framework trains LLMs to generate critiques of flawed outputs, validate corrections, and iteratively refine their ability to spot errors. Tang et al. (2025) report improvements of over 10% on critique benchmarks, showing that self-critique fosters scalable, autonomous oversight.

Certifying Judges. Naïve LLM judges are often inconsistent, rewarding stylistic cues over correctness (Bavaresco et al., 2024; Schroeder & Wood-Doughty, 2024).

Validity-Centered Evaluation. Certifying judges requires not only behavioral properties but also principled validity evidence. Salaudeen et al. (2025) introduce a framework that distinguishes between measurement, claims, and evidence, grounded in psychometrics. They emphasize construct, content, and criterion validity, and systematically enumerate threats to validity. This framework provides a blueprint for reporting and interpreting judge certification results, complementing identity/monotonicity checks with broader validity cards.

Robust Evaluation without Ground Truth. Standard judge queries can be brittle in the absence of ground truth, particularly under adversarial manipulation (Zhuo, 2024; Tong & Zhang, 2024). Robertson & Koyejo (2025) model the overseer as an agent and propose using information-theoretic, incentive-compatible queries instead of direct ratings. They introduce bounded f-divergence measures such as TVD-MI, which exhibit polynomial robustness to adversarial strategies, unlike unbounded KL-based criteria. This work highlights a pathway toward robust, certifiable judges when explicit ground truth is unavailable.

Single-judge certification vs. peer-prediction consensus. Information-elicitation approaches such as TVD-MI (Robertson & Koyejo, 2025) certify trust by asking multiple judges whether their outputs share information about the same source—fundamentally a consensus-based protocol over a judge population. Property certification is complementary: identity and monotonicity properties evaluate a single judge in isolation, with no agent-population dependency. This makes property certification deployable in settings where consensus is unavailable (single-judge inference, novel domains, sparse model panels), and our human-agreement assessment protocol amortizes the human-annotation cost that consensus mechanisms also incur.

Consensus aggregation vs. property certification. Denisov-Blanch et al. (2026) stress-test five inference-time aggregation strategies—majority vote, highest-confidence selection, confidence- and prediction-weighted votes, and the Surprisingly Popular algorithm—across multiple benchmarks (HLE, BoolQ, Com2Sense, and Predict-the-Future) and open-source models spanning roughly 4B to 235B parameters; even at $25\times$ the cost of single-sample inference no aggregator consistently beats a single sample in verifier-absent domains, and several amplify shared misconceptions. They trace the failure to structurally correlated LLM errors: in MATH plurality-wrong cases, the most common incorrect answer accounts for 65–87% of wrong responses, and in an out-of-distribution random-ASCII control with no ground truth, different models still agree at Cohen's κ as high as ≈ 0.35, indicating shared inductive biases rather than shared knowledge. A central diagnosis is that self-reported confidence and consensus track *what the crowd will say* more reliably than *what is true*, separating social prediction from truth verification, and the paper concludes that inference-time scaling improves truthfulness only when an external verifier is available. Our work sits on the verifier-present side of that boundary: instead of aggregating across a judge population, we evaluate a single judge against falsifiable properties (P1–P4) over Lean 4-checked gold artifacts and explicitly constructed bug/specification ladders, yielding a metric for assessing appropriate reliance with no inter-judge consensus dependency. For oracle-free extensions, TVD-MI-style cross-judge information should be used as a robustness diagnostic rather than as a truth proxy, side-stepping the consensus-as-truth failure mode they document; conversely, their negative results sharpen the case for property-based, single-judge certification whenever a verifier or reference checked against known outcomes exists.

Specializing judges to a test set vs. certifying judges across configurations. Finkelstein et al. (2025) propose the *Specialist* method, which deliberately abandons cross-test-set generalization and instead overfits a prompted LLM autorater to a single fixed benchmark. For each test example,

in-context demonstrations are drawn from historical human ratings of the *same source*, yielding 54–119% character-level F1 gains over XCOMET on the WMT'23/'24 MQM tasks. Their work and ours share a common motivation—amortizing the expensive part of human evaluation rather than re-annotating from scratch—but amortize along orthogonal axes. Specialist amortizes *within* a test set: it reuses many historical ratings (under a pseudo-SxS, single-rater-per-input regime) as inference-time demonstrations, so the judge must be re-specialized for each new benchmark and, as the authors show, to the individual rater. Property certification instead amortizes *across judge configurations*: a one-time human-agreement assessment panel relates falsifiable behavioral properties (P1–P4) to human agreement, after which any new $(\text{model}, \text{prompt}, n_{\text{repeats}}, \text{aggregation})$ tuple is screened by fully automated perturbation checks, with no human-vs-gold labels required on the perturbed variants. The two approaches are complementary: Specialist still consumes human ratings at scoring time and presumes a curated pool of same-source annotations, whereas our property-certification protocol targets the setting where new judges proliferate faster than any per-test-set rating pool can be maintained. Notably, their finding that a Specialist autorater specializes to a *single* rater echoes our own concern with human-pass dispersion; both works suggest single-rater ground truth is an imperfect ceiling, motivating the multi-rater held-out protocol we defer to future work.

Formal Methods as Oversight. Proof assistants provide binary, kernel-level guarantees. Zhou et al. (2024) autoformalize math solutions into Isabelle/HOL and show that filtering via proof checking outperforms majority voting by 12% on GSM8K. Liu et al. (2025) extend this approach with SAFE, which verifies every intermediate step in Lean 4 and introduces the FormalStep benchmark, reducing hallucinations and enabling fine-grained oversight.

Agentic Iterative Pipelines. Ospanov et al. (2025) integrate Lean feedback into recursive repair cycles, boosting general-purpose LLM proof success from 3–7% to over 40% on miniF2F, and achieving 75% with fewer than 1000 samples. Miranda et al. (2026) extend this paradigm to program verification, showing that self-debugging agents quadruple success compared to one-shot prompting. Incorporating an LLM judge improved completeness on security tasks but required tuning to avoid regressions on simpler cases.

Benchmarks for Judge Reliability. Benchmarks such as VeriBench and FormalStep quantify judge reliability across compilation, proof success, and specification quality. Miranda et al. (2026) report normalized specification scores of 0.470 for one-shot prompting versus 0.615 for self-debugging agents. Outside formal mathematics, MT-Bench and G-Eval provide complementary judge-evaluation settings using pairwise preferences and rubric-guided scoring, underscoring the need for human-agreement analysis and validity reporting (Zheng et al., 2023; Liu et al., 2023). Together with validity frameworks and robust information-theoretic scoring, these benchmarks advance the agenda of certifying LLM judges for formal verification.

Evaluating Autoformalization with Reference-Free Judges. In a complementary line of work, Zhang et al. (2025) propose an ensemble of LLM judges for the autoformalization task, using a fine-grained taxonomy to score formalizations without a gold-standard reference. The primary strength of this approach is its reference-free nature, offering a scalable evaluation pathway when gold standards are unavailable or prohibitively expensive to create. However, the work's motivation for abandoning gold-standard comparison is based on limited evidence: a small, non-random sample of "potentially incorrect" formalizations pre-selected by an LLM, whose own reliability is uncertified. Furthermore, the framework validates its judge by correlating with human preference scores—a subjective proxy—rather than the more objective, machine-checkable goal of logical equivalence to a known-correct specification. Critically, even if minor errors exist in gold-standard datasets, recent work has shown that such noise rarely alters the relative ranking of models, as errors are typically correlated across systems (Salaudeen & Hardt, 2024). This suggests that a small number of flawed examples does not invalidate the utility of an entire benchmark for comparative evaluation. Our work, in contrast, proceeds from gold references that Lean kernel-check relative to explicitly stated predicates, and emphasizes *development-panel evidence* for pairwise judge scores—the preferred rubric-aligned target when verifier-backed references exist.

B.2 HCI FOR SOLVING SCALABLE OVERSIGHT

From more evaluation to *human-centered* oversight. HCI reframes scalable oversight as a problem of matching reliance to demonstrated reliability: design interactions that let people verify, steer, and place *appropriate* trust in LLM outputs rather than maximizing raw trust (Bach et al., 2022). Interface choices (e.g., citations, explanations, confidence cues) can substantially shift user trust—sometimes in misleading ways—so the goal is to expose process and embed lightweight verification instead of presenting one-shot answers (Ding et al., 2025). User studies in AI pair-programming similarly surface needs to communicate model performance, provide control over output style/constraints, and expose uncertainty to avoid over-reliance (Wang et al., 2024b).

Hybrid human↔AI evaluation. HCI work operationalizes scalable evaluation by combining expert/lay judgments with automated judges. Humans annotate failures or verify slices of output; systems then train or condition LLM-based "reference evaluators" on those signals to scale up assessment while staying anchored to what users value (accuracy, safety, usefulness). This human-in-the-loop pattern complements LLM-as-judge approaches—whose reliability varies by task and setup (Bavaresco et al., 2024; Schroeder & Wood-Doughty, 2024)—and is especially relevant when pass@k or single-shot metrics are brittle.

Interfaces that *operationalize* oversight. HCI systems instantiate oversight as interaction loops that make correctness checkable *during* generation: (i) *test-driven code generation* (TiCoder) elicits yes/no judgments about candidate unit tests before showing code; a few interactions reduced cognitive effort and lifted pass@1 by $\sim$45.7% on coding benchmarks (Fakhoury et al., 2024); (ii) *explain→correct loops* (IntelliExplain) pair each snippet with a natural-language rationale that users can critique, improving novice task success by $\sim$11–25% (Yan et al., 2024); (iii) *pair debugging* (HypoCompass) has humans hypothesize faults while the LLM attempts fixes, yielding $\sim$12% learning gains (Ma et al., 2024); (iv) *multi-step pipelines* (PromptChainer) expose intermediate steps (e.g., tests $\rightarrow$ verification $\rightarrow$ proof attempt), enabling user intervention where failures arise (Wu et al., 2022); and (v) *source-grounded verification* (SymGen) highlights claim-level spans backed by exact sources and leaves unsupported text unhighlighted, speeding verification by $\sim$20% and raising user confidence (Hennigen et al., 2024). Collectively, these patterns shift trust from fluency to *verifiable progress*.

Applying HCI patterns to formal specs/theorems. Formal kernels (Lean/Isabelle) already deliver binary checks; HCI contributes the *workflow glue* that helps users evaluate *which* theorem/spec to prove and whether it is complete. Two actionable patterns: (1) *counterfactual spec probes*—UI toggles that remove a postcondition or inject a known bug and display the impact on tests/judge score—help detect missing clauses or overfitting; and (2) *stepwise proof oversight*—surfacing intermediate goals and marking unverified suggestions—keeps the human in control of proof search. In our setting, these patterns interface naturally with a graded, semantics-sensitive evaluator.

B.3 VERICODE

Bursuc et al. (2025) presumes well-formed formal specifications and evaluates end-to-end verification under those specs (in Dafny, Verus/Rust, or Lean) reporting verified pass rates. In practice, impactful problems often lack both complete specifications and ground-truth answers. Our VERIBENCH program targets this gap by challenging the *generation* of specifications (from code, tests and documentation) and then *grading* their quality prior to proof search using a Lean-aware LLM judge that we validate via falsifiable principles/properties (Identity, Monotonicity, and Prompt Stability/Invariance), addressing known pitfalls of LLM-as-a-judge such as bias and prompt sensitivity (Zheng et al., 2023; Bavaresco et al., 2024; Schroeder & Wood-Doughty, 2024). However, our framework has only been tested with a judge with gold references files with specifications but we will explore metrics/methods that don't require this like TVD-MI in (Robertson & Koyejo, 2025).

Beyond spec-first settings, the Unsolved Questions (UQ) paradigm evaluates models on difficult, real-world problems that have neither ground-truth answers nor formal specifications; instead, it relies on validator pipelines and community verification to track progress without oracles Nie et al. (2025). Taken together, these lines of work shift evaluation from "solve what is already specified" to "specify, validate, then solve": vericoding provides end-to-end rigor under given specs, VERIBENCH adds specification synthesis with a property-checked semantic judge, and UQ extends evaluation to

specless, answerless settings that require validator-assisted screening. Our contribution is a property-checked, semantics-sensitive judge that enables scalable oversight when the specification must be proposed rather than merely proved. Because specification generation often lacks a unique ground truth, we will extend our oracle-free property-assessment protocol with UQ-style validators, adding cycle-consistency tests (spec to code/docs) alongside our bug and ablation ladders, and enabling abstention when validators disagree to prioritize precision over coverage.

C FORMAL VERIFICATION CONTEXT AND TOOLING

Formal methods as scalable oversight. A central concern with LLM-assisted research is that models produce text faster than humans can audit, so trust in the output can erode even when the surface looks correct. The concern treats verification as a single unstructured activity: one human eye against a wall of generated tokens. With a proof assistant, the activity decomposes. The Lean kernel deterministically checks every step of a proof, so what the human must verify is not the proof but a much smaller artifact—the formal statement, its definitions, and its assumptions—and whether they faithfully express the informal claim. This is the structural move that makes oversight scalable: the verifier checks the answer, while the human checks the question. The premise is timely: autoformalization is improving, language-model-assisted proof search has become usable rather than a demonstration, and the surrounding tooling (`mathlib4`, Lean's metaprogramming layer, Pantograph, Aesop) has matured to the point where formal verification of nontrivial claims is no longer a niche enterprise. This view sharpens, rather than dissolves, the concerns motivating human oversight: shallow reading and unjustified confidence remain failure modes, but the target of human understanding shifts from hundreds of lines of generated argument to a Lean statement together with its definitions and assumptions—a much smaller object, but the one that actually encodes intent. Theoretical limits remain—Rice's theorem, the halting problem, and Gödelian incompleteness bound what any countable formal system can decide—but they constrain what the verifier can settle, not the central oversight claim that humans are best deployed on *asking the right question*. For complementary perspectives on verifier-present oversight and HCI patterns for exposing judge evidence, see the discussion in Section B.

Why Lean 4? Lean 4 is a fully-fledged programming language and lets VeriBench contain full runnable code and machine-checked proofs in the same dependently-typed language. Ahead-of-time compilation with the Lake toolchain produces fast native binaries, while first-class `Task` primitives, async I/O, and a thread-pool scheduler enable genuine concurrent programs inside the prover. Lean's C-level foreign-function interface (FFI) lets those binaries call out to high-performance libraries when needed. On the proof side, Lean ships a powerful metaprogramming system written in Lean itself, giving researchers a programming interface to access its internals. This led to the creation of tools such as Pantograph and Aesop. Finally, the community-maintained `mathlib4` and `std4` libraries supply thousands of reusable theorems and data structures, and they are expanding quickly thanks to an active Zulip and GitHub ecosystem. Lean also has a work-in-progress program verification library, CSLib, which has recently seen rapid growth. Lean 4 is backed by an unusually vibrant open-source community: hundreds of contributors refine `mathlib4` on GitHub each week, the public Zulip sees expert discussion around the clock, and even Fields-Medalist Terence Tao has chosen Lean to formalize portions of his current research—clear testimony to the ecosystem's accessibility and intellectual depth. Unlike the unit tests, fuzzers, and code sanitation pipelines common in industry—tools that sample inputs or rely on heuristics—Lean 4 supplies machine-checked proofs that a property holds for *all* executions. Its dependent type system can encode deep invariants (e.g., length-indexed arrays, bounded integers), so programs that violate them fail to compile, eliminating whole classes of bugs such as buffer overflows or integer wrap-around. Code, specification, and proof reside in the same file and are compiled by Lake into the shipped native binary, preventing the drift that arises when verification artifacts live outside the build. In short, Lean turns informal "best-effort" checks into formal, end-to-end guarantees without sacrificing performance or interoperability.

Lean 4's limitations. Lean's toolchain is still younger than Coq's or Isabelle's, making its standard libraries and automation smaller, and thus some formalizations demand extra groundwork. Lean's runtime has not been stress-tested at the scale of mainstream systems languages, meaning large-scale or safety-critical deployments may require additional vetting. Lean tactics could contain bugs and fail to typecheck on different environments. Acknowledging these gaps clarifies that VeriBench chooses Lean 4 for its unique unified programming-plus-proving model and modern automation hooks, not

because it already matches the decades-old industrial maturity of older theorem provers. *Static vs. runtime caveat:* Lean's guarantees are *static* and specification-relative: they certify the Lean program meets the stated properties, but they do not by themselves detect emergent runtime faults outside the model (e.g., I/O (Input/Output) failures, environment misconfiguration, FFI (Foreign Function Interface) unsoundness, resource exhaustion, or undefined behavior in linked C code). Such behaviors must either be modeled and proved, or mitigated with production safeguards (monitoring, sandboxing, input validation).

Lean 4 versus Dafny. Unlike Dafny, whose verifier translates each program into the Boogie intermediate language and then discharges first-order verification conditions with an SMT solver such as Z3, Lean 4 reasons natively in a dependently-typed calculus. Because Lean 4's types can mention run-time data and the very same source code is ahead-of-time compiled to a native binary via Lake, we can both state and prove value-indexed, higher-order properties (e.g., length-indexed vectorsand run the verified program itself—an end-to-end, fidelity that Dafny's SMT-centred, Boogie-to-Z3 workflow cannot natively match. This gives Lean the expressive power to specify and prove higher-order, data-dependent properties—precisely the kind of semantic guarantees VeriBench seeks to test—while still yielding runnable binaries compiled by the same toolchain. Dafny's SMT-centric workflow offers impressive push-button automation for imperative code but cannot natively encode the richer specifications (e.g., length-indexed vectors, algebraic invariants) that Lean handles directly.

Lean 4 versus TLA+. TLA+ excels at high-level specification of concurrent and distributed protocols, with correctness checked by the TLC model-checker and the TLAPS proof system that dispatches first-order obligations to external provers. However, TLA+ specifications are not executable programs; a separate implementation step is required, and state-space explosion can limit model-checking scalability. VeriBench instead needs a prover where the specification, proof, and runnable code live in the same language. Lean 4's dependently-typed core lets us capture fine-grained, data-dependent invariants and then compile the very same artifacts to fast native binaries—capabilities outside TLA+'s scope.

Lean 4 versus Verus. Verus is a verification framework for Rust programs using SMT solvers. A user state intermediate goals and an SMT backend fills in the intermediate steps. Compared to Verus, users have the option to use or not use SMT Solver-based tactics in Lean. This enables more fine-grained control and steering of the proof process. Verus also has a much larger kernel compared to Lean, and is more susceptible to soundness problem than Lean whose kernel is small.

D Future Work (cont.)

While (Robertson & Koyejo, 2025) study demonstrates that total-variation-based mutual information (TVD-MI) provides robust, gaming-resistant evaluation without ground truth, many settings —such as VeriBench— do provide gold specifications in Lean 4. A promising direction is to develop hybrid protocols that combine TVD-MI's black-box robustness with supervised ground-truth checks (compile success, unit tests, post-condition proofs, and equivalence theorems). By fitting mappings from MI scores to known gold outcomes, we can create composite metrics that preserve gaming-resistance while offering probabilities of correctness whose predictions match observed frequencies where supervision exists. This opens the door to semi-supervised label propagation, active selection of items to prove, and cross-system experiments in domains with heterogeneous supervision. Such an integration would extend Robertson's information-theoretic framework into a practical, specification-aware oversight system that leverages the best of both unsupervised robustness and supervised precision.

E Why Judge Scores Are Specification-Relative

It is tempting to interpret a perfect score from our LLM judge as a guarantee of absolute program correctness. However, this is fundamentally impossible due to established limits from computability theory. The obstacle is not a lack of ingenuity in our benchmark design or judge prompting, but an established negative result.

Rice's Theorem. The core limitation stems from Rice's theorem, which states that any non-trivial, semantic property of programs in a Turing-complete language is undecidable.

> **Rice's Theorem (Informal):** Let $\mathcal{S}$ be any property of a program's input-output behavior (e.g., "the program always terminates" or "the output is always sorted"). If $\mathcal{S}$ is non-trivial (meaning some programs have it and some don't), then no algorithm can exist that decides for *every* program whether it satisfies $\mathcal{S}$.

Impact on the Gold Standard and the Judge. Rice's theorem directly implies that the gold reference files used by our judge are necessarily incomplete. If a recursively enumerable gold standard could capture *every* true semantic property of a program, one could use it to build a decision procedure that contradicts the theorem. Therefore, our gold standards, while rigorous, represent a curated but fundamentally finite set of important specifications.

This leads to a critical limitation for interpreting our artifact scores: evaluations are **specification-relative**. A maximal score denotes close agreement with *our* audited gold formalization and rubric-conditioned human ratings, not a universal guarantee beyond that reference.

F Unit Tests as Ground Theorems: A Definition We Do Not Adopt

Section 3 follows common formal-methods usage: the formal specification consists of theorem statements about the implementation, and unit tests are not part of it. We record a competing view that we do not adopt. In Lean 4, a unit test is often itself a closed theorem. The gold `myAdd` artifact states its tests as, e.g., `example : myAdd 1 2 = 3 := by native_decide`, an anonymous theorem with no free variables. On this view, a unit test specifies the implementation at one concrete input, and specifications range from such ground instances to universally quantified properties such as $\forall n$, `myAdd` $n\ 0 = n$. The choice changes terminology, not our measurements. The P3 ladder removes theorem statements and unit tests alike; under the adopted definition, only the removed theorems count as missing specification.

G Judge Prompt Templates

For reproducibility, we include the main judge prompt templates used in the paper's principal judge panels. The primary Codex / Opus / Qwen comparisons use the prompt variants `P1_BASELINE`, `P4_CHECKLIST`, and `P8_STRICT_B`; the BAD_STRINGY lexical baseline is included as a negative control.

P1_BASELINE

```
You are an expert in Lean 4 formal verification.
Score how well the GENERATED THEOREM captures the correctness properties
of the REFERENCE IMPLEMENTATION. Score from 0.0 to 1.0.

- 1.0: Semantically equivalent to what the reference guarantees
- 0.7-0.9: Captures most properties, misses minor ones
- 0.4-0.6: Captures some properties, has meaningful gaps
- 0.1-0.3: Technically valid Lean but misses core properties
- 0.0: Trivially true (e.g., `True`) or incorrect

REFERENCE IMPLEMENTATION:
{reference_impl}

GENERATED THEOREM:
{generated_theorem}

Respond ONLY with valid JSON, no markdown:
{"score": <float 0.0-1.0>,
 "reasoning": "<one sentence max>"}
```

P4_CHECKLIST

```
Evaluate this in order:
1) What properties does REFERENCE IMPLEMENTATION guarantee?
2) Which are preserved by GENERATED THEOREM?
3) Are there missing core guarantees or trivialization?
4) Map to score in [0.0, 1.0]:
   1.0 equivalent; 0.7-0.9 minor gaps;
   0.4-0.6 meaningful gaps; 0.1-0.3 weak;
   0.0 trivial/incorrect.
```

```
REFERENCE IMPLEMENTATION:
{reference_impl}

GENERATED THEOREM:
{generated_theorem}

Return JSON only:
{"score": <float 0.0-1.0>,
 "reasoning": "<one sentence>"}
```

P8_STRICT_B

```
Evaluate conservatively. If uncertain, choose the lower score.
Use [0.0,1.0]:
1.0 equivalent semantics
0.7-0.9 mostly right, minor misses
0.4-0.6 substantial omissions
0.1-0.3 weak relation
0.0 trivial/incorrect

REFERENCE IMPLEMENTATION:
{reference_impl}

GENERATED THEOREM:
{generated_theorem}

Return valid JSON only:
{"score": <float>,
 "reasoning": "<one sentence>"}
```

BAD_STRINGY negative control

```
You are a text similarity grader.
Compare generated theorem text to the reference implementation text.
Do NOT evaluate semantic correctness or theorem meaning.
Score ONLY based on superficial lexical/format overlap:
- high score if tokens/phrases look similar
- low score if wording/structure differs
Reference implementation:
{reference_impl}

Generated theorem:
{generated_theorem}

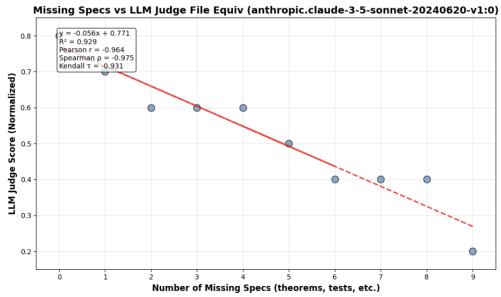
Return ONLY a JSON object:
{"score": <float 0.0 to 1.0>,
 "reasoning": "<short reason>"}
```

H JUDGE CERTIFICATION: ADDITIONAL PLOTS

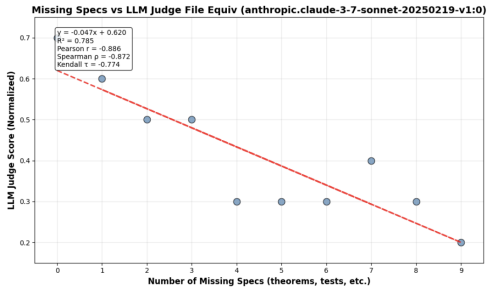
We report raw scatterplots used during certification of the LLM judge. These plots visualize the judge's response to controlled perturbations: (P1) identity/reflexivity, (P2) cumulative bug injection, and (P3) removal of theorem statements and unit tests. Figures A.1–A.3 show that scores drop as artifacts worsen; the bug plot is noisier with an early prompt, which motivated the simplified rubric used in the main experiments. P4 (prompt stability/invariance) is reported in the main text with dedicated aggregate plots.

Protocol variants. The scatterplots below are "sanity-check" runs with the simplest prompts to stress visible monotonic behavior. In the bug study (Fig. 4) the earlier, theorem-equivalence prompt produced a noisier trend; this prompted a simplified, rubric-first prompt that we use for all reported results in the main paper, yielding the stronger correlations cited in Fig. 1. For missing-specs (Fig. 3), both Claude 3.5 and 3.7 show clear negative trends; minor non-monotonicity in 3.7 disappears with the finalized rubric.

We note that, for the experiment shown in Figure 3, a more complex judge (Claude 3.7) does not satisfy the property that the LLM Judge Score should monotonically decrease with the number of missing specs. Indeed, the Judge File Equivalence score was not monotonic, even for Claude 3.5, leading us to simplify the rubric used in our main paper.



(a) Claude 3.5 Sonnet v1: strong monotonic decrease ($y = -0.056x + 0.771$, $R^2 = 0.929$, Pearson $r = -0.964$, Spearman $\rho = -0.975$).



(b) Claude 3.7 Sonnet v1: clear negative trend with minor non-monotonicity ($y = -0.047x + 0.620$, $R^2 = 0.785$, Pearson $r = -0.886$, Spearman $\rho = -0.872$).

Figure 3: **(P3) Monotonicity w.r.t. missing specifications.** Candidate scores decrease as more theorems/tests are withheld. We retain both model variants to show robustness of the trend; 3.7's small non-monotonicity motivated the simplified rubric used in the main results.

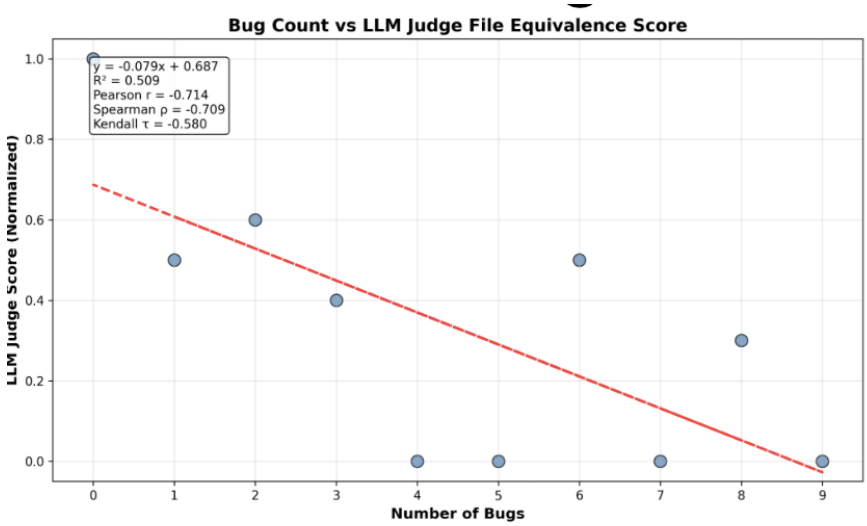


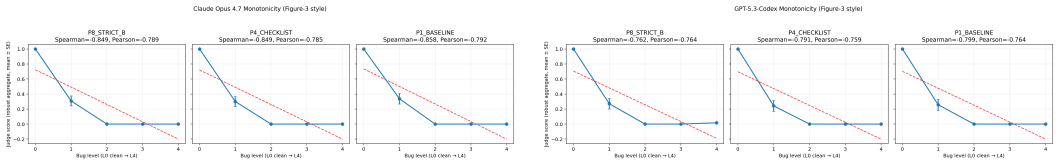
Figure 4: **(P2) Non-Monotonicity w.r.t. cumulative bugs (Claude 3.5 Sonnet).** Scores drop as more semantic bugs are injected (Pearson $r = -0.714$, Spearman $\rho = -0.709$, $R^2 = 0.509$); the higher noise with the initial "theorem-equivalence" prompt led us to simplify the rubric used for the final judge in the main paper.

I Development-Panel Human Alignment Appendix

This appendix reports full visual diagnostics for the final robust development-panel human-alignment analyses under three label views: pass1, pass2, and avg.

I.1 Naive Codex-Style Blind Baseline

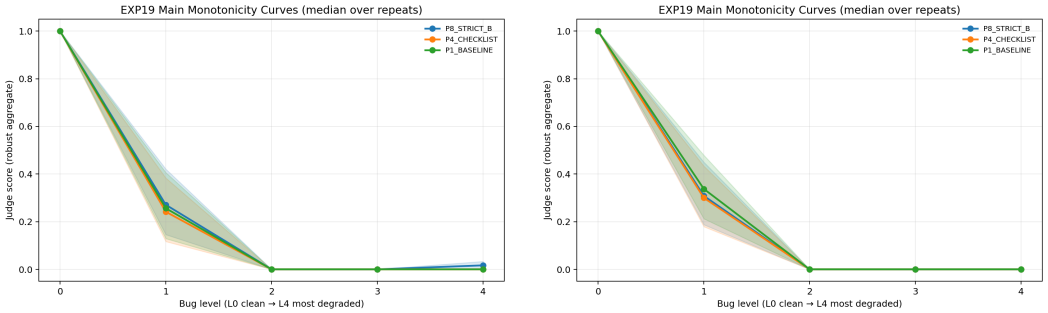
To contextualize TI-ranked judge performance, we evaluate a naive blind baseline on the same pair set ($n = 5$ overlap), using a single-pass, surface-form-oriented Codex-style rubric without property-based tuning constraints (no tuning for monotonicity/stability). The baseline tracks pass2 labels somewhat but remains substantially weaker and less consistent than the top TI-ranked judges, especially on pass1 and in absolute error.



(a) Claude Opus 4.7 Figure-3-style monotonicity (mean ± SE).

(b) GPT-5.3-Codex Figure-3-style monotonicity (mean ± SE).

Figure 5: **Model-matched Figure-3-style monotonicity diagnostics.** Both judge families show strong monotone score degradation as bug intensity increases, under the same low-variance plotting protocol.



(a) Codex low-variance monotonicity curves.

(b) Opus 4.7 low-variance monotonicity curves.

Figure 6: **Low-variance monotonicity trends across model families.** Both judge families show strong non-increasing behavior as bug intensity increases, under robust repeat aggregation.

I.2 Human-Human Agreement Ceiling Context

Development-panel judge-human Spearman values should be interpreted relative to the reliability of human labels themselves. Across the two human annotation passes on the aligned set ($n = 75$), inter-pass agreement is moderate (Spearman 0.249, Pearson 0.375, Kendall 0.221, QWK 0.372). Thus, development-panel judge-human Spearman values in the 0.48–0.56 range are non-trivial and should be interpreted as approaching the available annotation ceiling in this dataset rather than as "low absolute" values.

I.3 Cross-Model Validation Status (Full vs Preliminary)

To avoid over-claiming, we separate full-scale cross-model regressions (n=8 judge tuples on the ∼72–75 overlapping aligned rows for each labeling view) from narrower exploratory protocol plots. **Anthropic Claude Opus (4.x)** $\mathrm{TI}_{\mathrm{var}}$ vs. development-panel human Spearman is a full-panel run of the same form as GPT-5.3-Codex and Qwen3-Coder in Figure 2: eight judge configurations (four Opus × prompts plus cross-family `gpt-4o-mini`×two prompts plus edit/random controls), repeated-judge aggregation, identical label sets.

Early **Claude 3.x Sonnet** scatterplots reported in Appendix H were small-file protocol iterations used mainly to stabilize prompts and ladders; those plots illustrate qualitative monotonic behavior but are *not* the statistic behind Table 3.

I.4 Leanstral Negative-Control Reaggregation (AM/GM/Variance-Aware)

To stress-test whether index choice alone can "rescue" a weak judge family, we re-aggregate the same Leanstral judge outputs under three indices:

$$\mathrm{AM} = \text{cert_index_v2} \qquad \text{(arithmetic baseline)}$$

$$\mathrm{GM} = \mathrm{mono}^{0.4} \cdot \mathrm{stab}^{0.6}$$

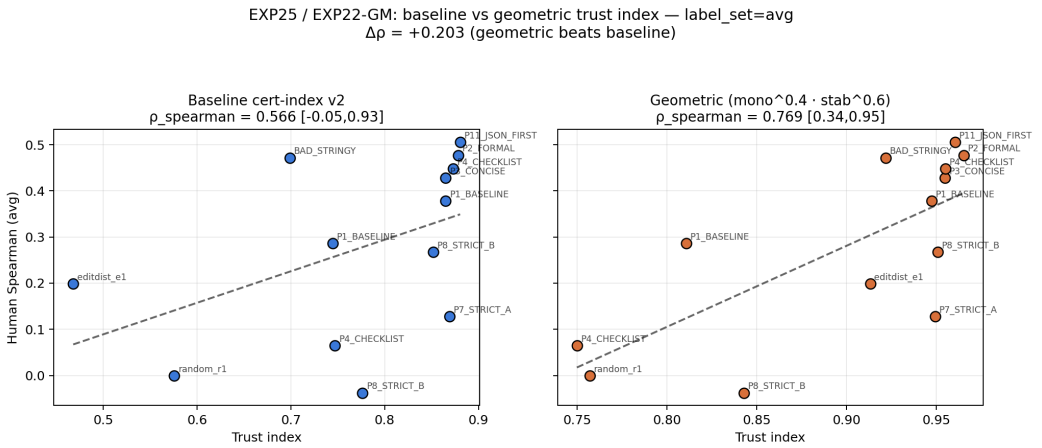
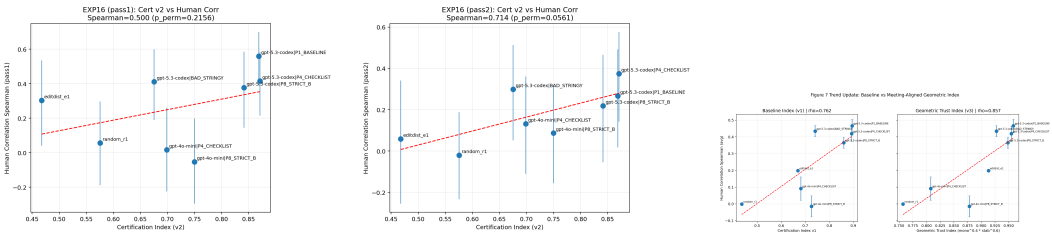


Figure 7: **Expanded 13-judge panel trend (avg development-panel labels).** Geometric trust index retains strong alignment with human correlation at larger judge-panel size ($\rho = 0.769$); pass-specific values are pass1 $\rho = 0.764$, pass2 $\rho = 0.753$.



(a) Judge-level trend (pass1) (b) Judge-level trend (pass2) (c) Judge-level trend (avg)

Figure 8: **Judge-level $\mathrm{TI}_{\mathrm{var}}$-to-human trend across development-panel label sets.** Across pass1, pass2, and averaged labels, higher certification strength is associated with stronger human agreement.

$$\mathrm{TI}_{\mathrm{var}} = \exp\Big(\tfrac{1}{3}\log C_f + \tfrac{1}{3}\log(1-\widetilde{\mathrm{VarCorr}}) + \tfrac{1}{3}\log(1-\widetilde{\mathrm{VarScore}})\Big)$$

where C_f is the mean per-file property-correlation term, and $\widetilde{\mathrm{VarCorr}}, \widetilde{\mathrm{VarScore}}$ are normalized variance penalties (lower is better before $1-\cdot$). We report Spearman between index score and development-panel human Spearman across judges.

I.5 Index Definitions

For reference, the simple arithmetic baselines in the main text are

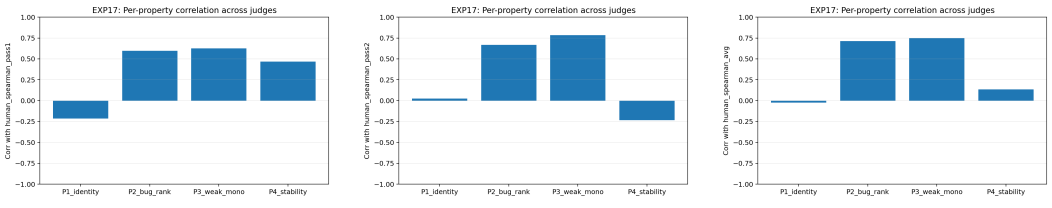
$$v1 = 0.30\,\text{ident} + 0.30\,\text{mono_rank} + 0.25\,\text{weak} + 0.15\,\text{stab},$$
$$v2 = 0.20\,\text{ident} + 0.20\,\text{mono_rank} + 0.15\,\text{weak} + 0.10\,\text{stab} + 0.20\,\text{anti} + 0.15\,\text{subtle},$$

where $\text{mono_rank} = (1-\rho_{\text{bug}})/2$ rescales bug-ladder Spearman so that larger is better, and $\text{stab} = 1 - \text{flip_rate}$.

For the variance-aware index, the monotonicity term is

$$C_f = \frac{1}{|\mathcal{F}|}\sum_{f\in\mathcal{F}} \mathrm{clip}\left(\frac{1-\rho_f}{2}, 0, 1\right),$$

where ρ_f is the Spearman correlation between bug level and judge score for file f. Thus C_f is the mean per-file bug-ladder monotonicity term before the variance penalties are applied.



(a) Per-property trend (pass1) (b) Per-property trend (pass2) (c) Per-property trend (avg)

Figure 9: **Property-wise development-panel human-alignment diagnostics.** P3 weak monotonicity has the strongest association across label sets; P2 is second; P1/P4 are weaker standalone signals at this scale.

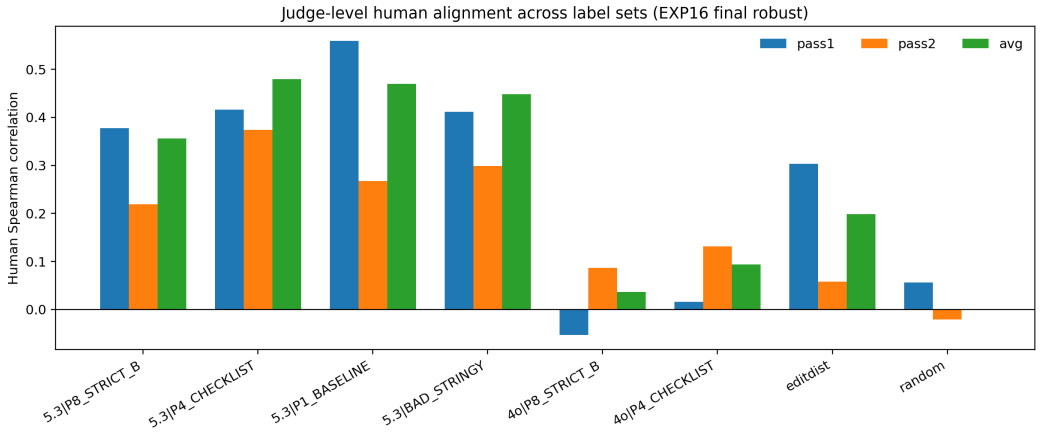


Figure 10: **Judge-wise human alignment across label passes.** Development-panel human Spearman across pass1, pass2, and averaged labels. GPT-5.3-Codex variants remain above weaker baselines on most label sets; pass-to-pass variation is visible and motivates reporting all three label sets.

I.6 Failure Modes Found by Certification (Not Hidden)

The certification framework is useful precisely because it identifies failures: **(i) alpha-renaming sensitivity** is the dominant identity/invariance gap, and **(ii) local bug-ladder reversals** occur despite overall negative monotone trends. We therefore report both strengths and weaknesses, and include alpha-ablation diagnostics. On trend-v1 Spearman (TI $\rightarrow$ development-panel human Spearman), excluding alpha variants changes pass1 $0.810 \rightarrow 0.690$, pass2 $0.810 \rightarrow 0.714$, avg $0.881 \rightarrow 0.786$, showing that alpha cases materially affect measured robustness.

I.7 Newest Results (Latest Rerun at Appendix Bottom)

To reduce judge-call noise, we reran the multi-judge development-panel trend with higher repeats and robust repeat aggregation: `JUDGE_REPEATS=9 REPEAT_AGG=median BOOTSTRAP_ITERS=10000 PERM_ITERS=20000`. This run writes to `results_exp16_final_robust_r9_median`.

We also generated a low-variance replot package from the same detail rows (`results_exp19_low_variance_replot`): a cleaner main monotonicity curve for core judges plus separate failure-mode plots for alpha sensitivity and local bug reversals. Reversal rates in this package are: GPT-5.3 P4/P1 $= 0.00$, GPT-5.3 P8 $= 0.167$, BAD_STRINGY $= 0.833$, GPT-4o variants $= 0.667$–0.750, edit-distance/random $= 1.00$.

Label Set	ρ_s	r	τ	MAE
pass1	0.171 ($p = 0.151$)	0.218 ($p = 0.0656$)	0.155 ($p = 0.144$)	1.049
pass2	0.388 ($p = 7.58 \times 10^{-4}$)	0.427 ($p = 1.87 \times 10^{-4}$)	0.343 ($p = 9.40 \times 10^{-4}$)	1.333

Table 9: **Naive blind baseline vs. development-panel labels (0–5 scale).** ρ_s: Spearman, r: Pearson, τ: Kendall; MAE is absolute error on the 0–5 scale. This baseline is included as a sanity-check negative control and is not used for final ranking claims.

Quantity	Value	Notes
Human-human Spearman (pass1 vs pass2)	0.249	95% CI = $[-0.015, 0.489]$
Human-human Pearson (pass1 vs pass2)	0.375	95% CI = $[0.076, 0.604]$
Human-human QWK (pass1 vs pass2)	0.372	95% CI = $[0.073, 0.581]$
Human-human MAE (pass1 vs pass2)	0.778	On aligned $n = 75$, 0–5 rubric
Human-human MSE (pass1 vs pass2)	1.139	On aligned $n = 75$, 0–5 rubric
JS divergence (pass1 vs pass2)	0.081 bits	Symmetric histogram divergence
KL(pass1‖pass2)	0.415 bits	Asymmetric divergence
Best development-panel Spearman (avg labels)	0.480	Best single judge on avg labels
Best development-panel Spearman (pass1 labels)	0.559	Best single judge on pass1 labels

Table 10: **Human-human agreement ceiling context.** Inter-pass human agreement is moderate, and inter-pass dispersion (MAE/divergence) is non-zero, so development-panel judge-human values near 0.5 are meaningful on this benchmark and should be interpreted with label-noise context.

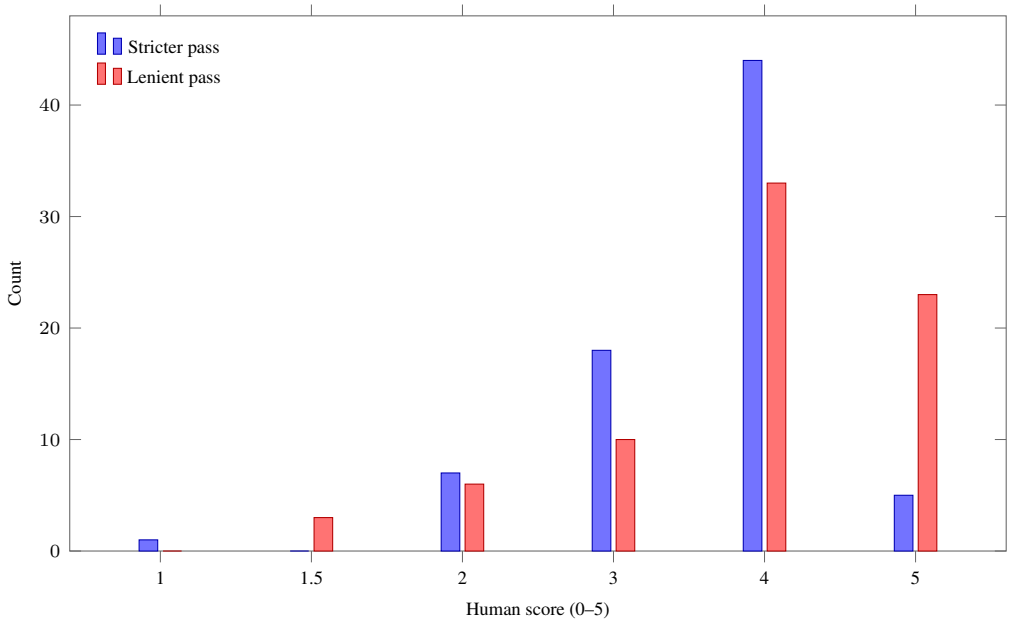


Figure 11: **Human-score dispersion across passes (aligned $n = 75$, 0–5 scale).** Score counts are computed on the full set of 75 aligned candidate–gold pairs shared across both passes. The lenient pass includes three 1.5 scores because duplicated legacy entries for one problem were averaged before plotting. The visible distribution shift is consistent with the non-zero MAE/MSE and divergence statistics in Table 10.

Judge setup	Scale	Main?	Role
GPT-5.3-Codex prompts (P8/P4/P1)	n=8	Yes	Primary judges
GPT-5.3-Codex BAD_STRINGY	n=8	Yes	Same-model stress
Qwen3-Coder prompts	n=8	Yes	Open-weights
Qwen3-Coder BAD_STRINGY	n=8	Yes	Open-weights stress
Claude Opus 4.x + controls	n=8	Yes	Third cross-model family
GPT-4o-mini prompts	n=8	Yes	Lightweight baseline
Edit-distance + random	n=8	Yes	Non-LLM controls
Claude 3 Sonnet (narrow App. slice)	Small	No	Prompt diagnostics

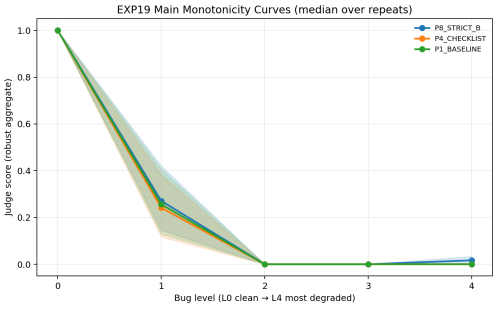
Table 11: **Cross-model validation status.** Main $\mathrm{TI}_{\mathrm{var}}$ claims use full eight-configuration panels on overlapping human-aligned rows; Sonnet exploratory figures are withheld from headline statistics.

Label Set	AM (v2)	GM	Variance-Aware $\mathrm{TI}_{\mathrm{var}}$
val (pass1)	-0.5952	-0.1429	-0.1190
test (pass2)	-0.1905	-0.2619	-0.0476
avg	-0.4286	-0.2143	-0.0952

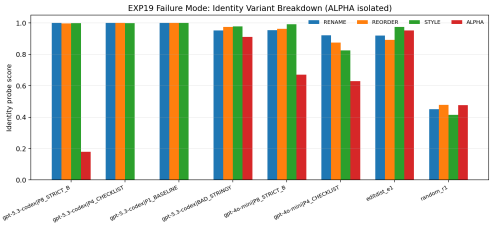
Table 12: **Leanstral remains low-alignment under multiple index constructions.** Re-aggregation improves over the arithmetic baseline in some splits, but all three settings remain near-zero or negative. This suggests the bottleneck is likely the underlying judge behavior for this ranking task (human-preference alignment), not only index form. One plausible hypothesis is that Leanstral is stronger for Lean proof editing/generation than for pairwise judging assessed against human ratings under this rubric; we treat this as a testable conjecture, not a final causal claim.

Label Set	v1 Spearman	v2 Spearman
pass1	0.762 ($p = 0.0372$, CI $[-0.139, 1.000]$)	0.524 ($p = 0.1959$, CI $[-0.400, 0.975]$)
pass2	0.810 ($p = 0.0222$, CI $[0.291, 1.000]$)	0.571 ($p = 0.1477$, CI $[-0.231, 0.975]$)
avg	0.762 ($p = 0.0402$, CI $[0.103, 1.000]$)	0.476 ($p = 0.2426$, CI $[-0.425, 1.000]$)

Table 13: **Latest rerun (R9 + median aggregation).** Increasing repeats and using median aggregation preserves the v1 trend signal while v2 remains more sensitive at this $n_{\text{judges}} = 8$ scale.



(a) Main low-variance monotonicity curves (core GPT-5.3 judges, robust repeat aggregation).



(b) Failure-mode isolation: identity variant breakdown with ALPHA shown separately.

Figure 12: **Low-variance replot package: cleaner main trend + explicit failure separation.** The left panel is the "main story" plot with lower visual variance; the right panel isolates alpha-renaming sensitivity rather than mixing it into the primary monotonicity figure.

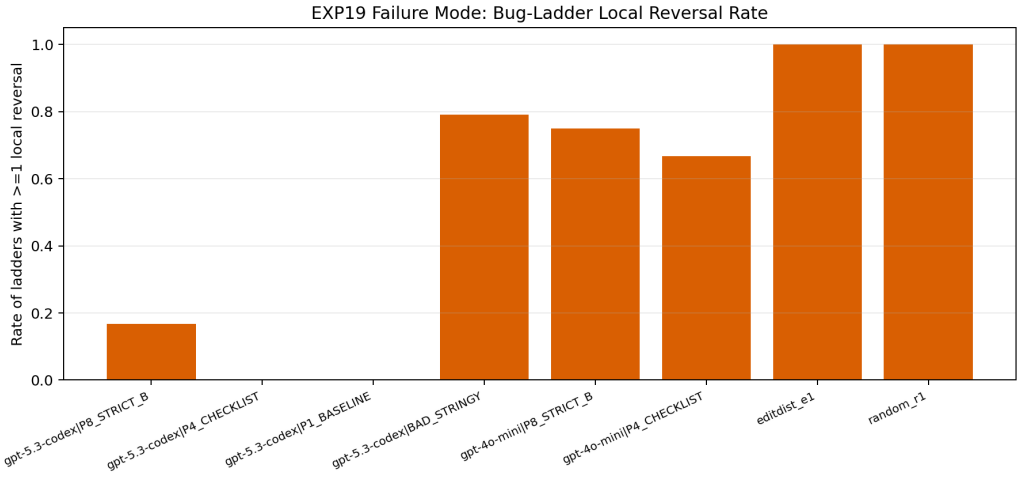


Figure 13: **Failure-mode plot: local bug-ladder reversal rates by judge.** This explicitly surfaces where monotonicity fails locally, separating robustness diagnostics from the primary trend plot.